Manimaran K

+91 7010723571 | manimaran230806@gmail.com | linkedin.com/in/manimaran-k | github.com/Manimaran-tech | manimaran.web.app

Summary

Third-year Computer Science undergraduate with production experience building **AI agent pipelines** (LangChain, LangGraph, Pydantic AI, n8n), ML systems (PyTorch, TensorFlow, PINNs, LSTM), and MLOps workflows (Docker, GitHub Actions, Firebase). Research Intern at **IIT Bombay FOSSEE**; authored **12+ merged PRs** into PyTorch-Ignite (4.9K GitHub stars). Hands-on with RAG pipelines, vector databases (ChromaDB, Pinecone), Hugging Face Transformers, and prompt engineering. Open to remote/hybrid AI internship roles.

Education

Rajalakshmi Institute of Technology — Chennai, India
B.E, Computer Science & Business Systems — GPA: 8.5/10 — *2024 – 2028*
- Relevant coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering

Velammal Vidyalaya (CBSE) — Chennai, India
Class XII: 93.4% | Class X: 88% — *2022 – 2024*

Experience & Open-Source Contributions

Research Intern — Nov 2024 – Feb 2025
FOSSEE, IIT Bombay — *Remote*
- Tested, debugged, and patched the open-source EDA tool **eSim** (Python), serving **10,000+ users globally**; all fixes delivered via Git pull requests reviewed and approved by IIT Bombay faculty.
- Authored **automated regression test suite** for the eSim 2.4 release, catching edge-case regressions missed by manual QA; documented root-cause analyses and runbook-style troubleshooting guides adopted into official project documentation.

Open-Source Contributor — 2024 – Present
PyTorch-Ignite & Gammapy — *Remote*
- **PyTorch-Ignite** (4,900+ GitHub stars): Authored **12+ merged PRs** implementing new evaluation metrics and leading module-wide code modernization across core metrics and handlers; all changes validated through GitHub Actions CI/CD pipelines.
- **Gammapy**: Refactored legacy code and removed deprecation warnings across a large multi-contributor scientific Python codebase with strict peer-review standards; earned **GitHub Pull Shark** achievement for sustained, high-quality contributions.

Projects

AutoReel | *LangChain, LangGraph, FastAPI, RAG, AI Agents, E2E Testing* | GitHub — 2025
- Engineered a **multi-stage agentic pipeline** (LangGraph orchestration): web scraping → RAG-based LLM text generation → voice synthesis → video rendering; end-to-end execution in under 3 minutes with zero manual intervention.
- Designed RESTful FastAPI backend with **ChromaDB vector store** for semantic retrieval; implemented full E2E automated test coverage across all pipeline stages using pytest and GitHub Actions.

YieldSense | *Firebase, Cloud Functions, CI/CD, ML Forecasting, Telegram* | GitHub — 2024
- Deployed and maintained a **live production application** on Firebase (Firestore, Auth, Cloud Functions, Hosting) with automated CI/CD pipelines achieving **100% uptime** over 3+ months.
- Built automated **nightly monitoring** with Telegram alerting via Cloud Functions, reducing mean time to detect failures to under 5 minutes; integrated Scikit-learn ML forecasting models with a live data visualization dashboard.

Satellite Re-entry Prediction via PINNs | *PyTorch, Physics-Informed Neural Networks* | GitHub — 2024
- Developed a **Physics-Informed Neural Network** in PyTorch for orbital decay prediction; designed a dual-objective loss function combining data-driven learning with physics constraints — achieved prediction error within **5% of known orbital mechanics baselines**.
- Demonstrated ability to validate ML model outputs against physical ground truth — directly applicable to scientific AI and MLOps evaluation pipelines.

SQL Data Analyst | *Docker, PostgreSQL, Python, Automated Grading* | GitHub — 2024
- Built a **containerized auto-grading platform** (Docker + PostgreSQL) validating SQL query outputs against a 5,000+ row, 8-table schema with planted edge cases — directly applicable to LLM output validation and MLOps testing pipelines.

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, YAML, Java, C, HTML/CSS

AI & LLM: LangChain, LangGraph, Pydantic AI, Swarm Agents, n8n, Prompt Engineering, RAG, Embeddings, Fine-tuning (LoRA)

ML & Data: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, LSTM, PINNs

Vector DBs: ChromaDB, Pinecone, FAISS

AI Agents & Tools: Claude Code, Codex CLI, GitHub Copilot, NIM NVIDIA, Antigravity

MLOps & DevOps: Docker, GitHub Actions, Firebase, Google Cloud, Git, CI/CD, MLflow (learning), Kubernetes (learning)

Frameworks: FastAPI, React.js, Next.js, Node.js

Practices: REST APIs, E2E Testing, Regression Testing, Microservices, Technical Documentation, Agile/Scrum